# 信 息 工 程 学 院

## 《大数据采集与清洗》实验报告

**学院**：信息工程学院　**班级**：数科 2102　**姓名**：陈　**学号**:202101　**成绩**:

## 一、实验题目

Kafka Python 的简单调用（Python 版）

## 二、实验目的

掌握 kafka-python 包的使用

## 三、实验平台

希冀平台

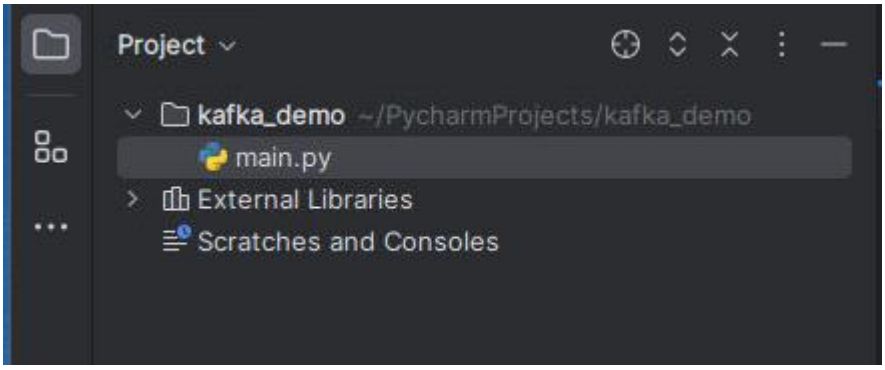
Kafka 3.5.1

## 四、实验过程及步骤

1. 启动 Kafka 环境

```
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# cd /usr/local/support/kafka_2.12-3.5.1
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/zookeeper-server-start.sh config/zookeeper.properties
[2023-11-25 01:13:30,540] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.quo
rum.QuorumPeerConfig)
[2023-11-25 01:13:30,541] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.a
pache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,544] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,545] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,545] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,545] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (o
rg.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,546] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 01:13:30,546] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 01:13:30,546] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 01:13:30,547] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.
zookeeper.server.quorum.QuorumPeerMain)
[2023-11-25 01:13:30,548] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2023-11-25 01:13:30,548] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.quo
rum.QuorumPeerConfig)
[2023-11-25 01:13:30,548] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.a
pache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,549] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 01:13:30,549] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
```

2. 使用 PyCharm 创建项目 kafka_demo

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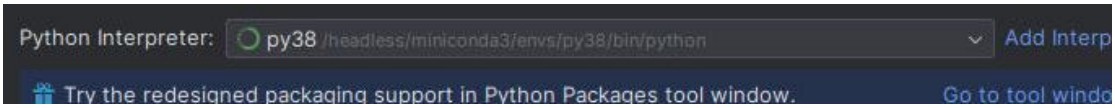
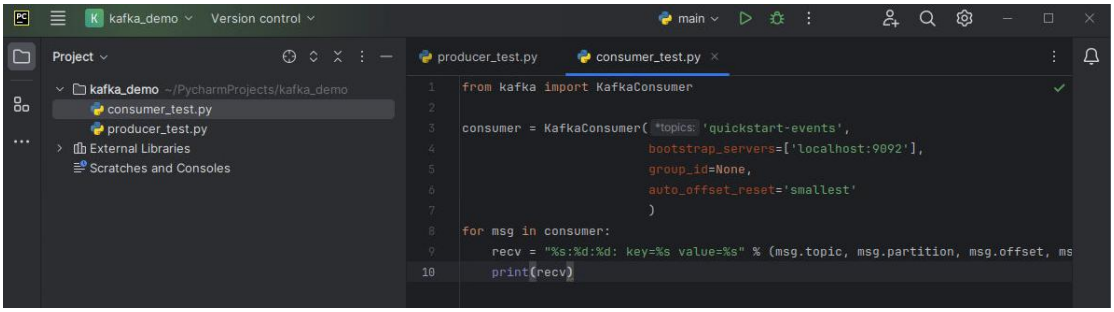
实验报告的内容与格式按任课教师的要求书写。

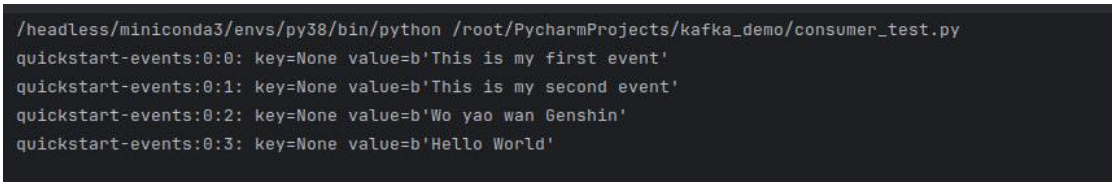
3. 添加解释器




4. 创建测试文件 producer_test.py，consumer_test.py

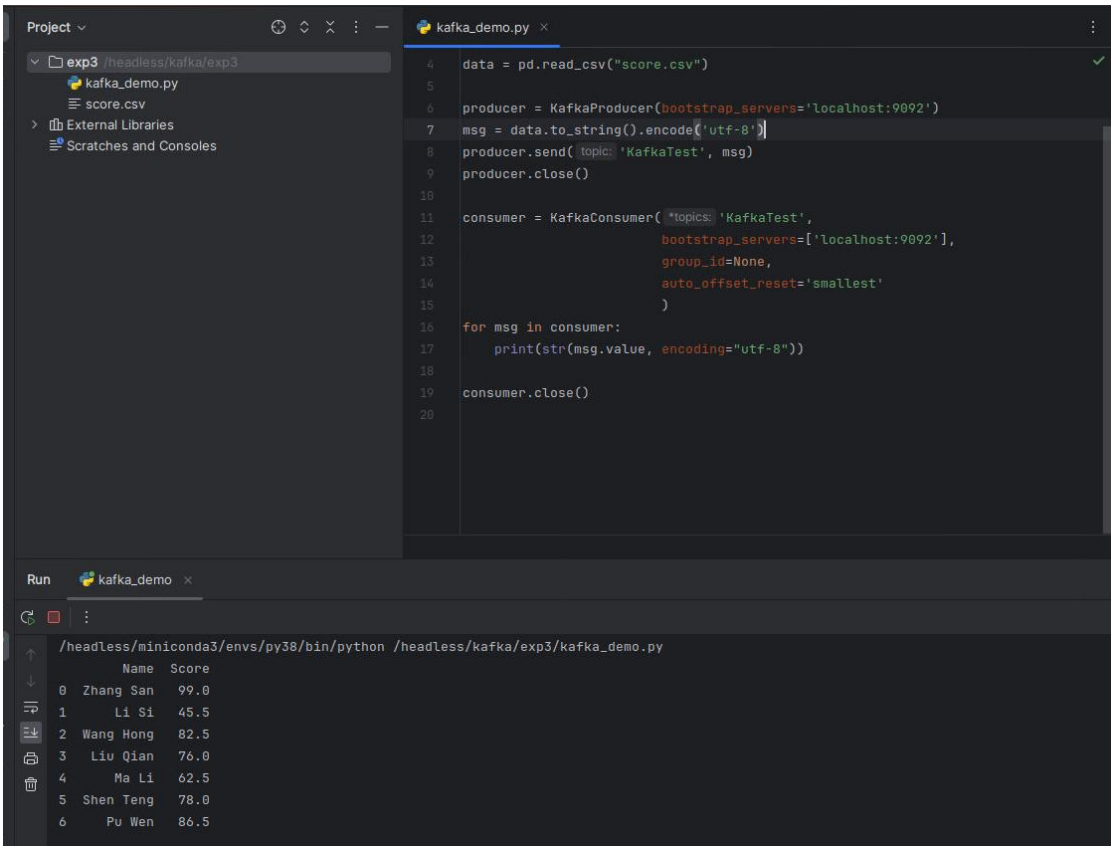



5. 先执行 producer_test.py 文件，再执行 consumer_test.py 文件，效果如下




实验任务

创建文件 kafka_demo.py 文件到/headless/kafka/exp3/目录下，要求编写生产者读取 score.csv 文件的所有内容，以 json 字符串发送到 Kafka，消费者接收到消息转为表格打印，效果如下



```
/headless/miniconda3/envs/py38/bin/python /headless/kafka/exp3/kafka_demo.py
         Name  Score
0  Zhang San   99.0
1      Li Si   45.5
2  Wang Hong   82.5
3   Liu Qian   76.0
4      Ma Li   62.5
5  Shen Teng   78.0
6     Pu Wen   86.5
```

```
from kafka import KafkaProducer, KafkaConsumer
import pandas as pd

data = pd.read_csv("score.csv")

producer = KafkaProducer(bootstrap_servers='localhost:9092')
msg = data.to_string().encode('utf-8')
producer.send('KafkaTest', msg)
producer.close()

consumer = KafkaConsumer('KafkaTest',
                         bootstrap_servers=['localhost:9092'],
                         group_id=None,
                         auto_offset_reset='smallest'
                         )
for msg in consumer:
    print(str(msg.value, encoding="utf-8"))

consumer.close()
```